SDC Learn AI: Course Generation

Final plan: one document for the whole feature. Skill Development Council Karachi. 13 October 2026.

Item	Detail
What this is	The single plan for building AI course generation in SDC Learn: what it does, how the user uses it, how it is built, in which order, and how we know it works.
Status	Draft for the team. Nothing is built yet. Building starts when the team accepts this plan (section 13 lists the choices to confirm).
What it replaces	The earlier BRD v0.1, BRD v0.2, the Visual Overview and the seven review documents. Their content is merged here. Where they disagreed, this document picks one answer.
What is left out on purpose	Security checks and platform hardening. This document covers the feature only.

1. The idea in short

Today a coordinator builds each course by hand: one form for the course, then one form for every session. A course with 24 sessions means filling the session form 24 times.

With the course generator, the coordinator writes a short description of the course (the **brief**). The AI writes a first draft of the course. The coordinator checks it, changes what they want, and saves it as a **draft course**.

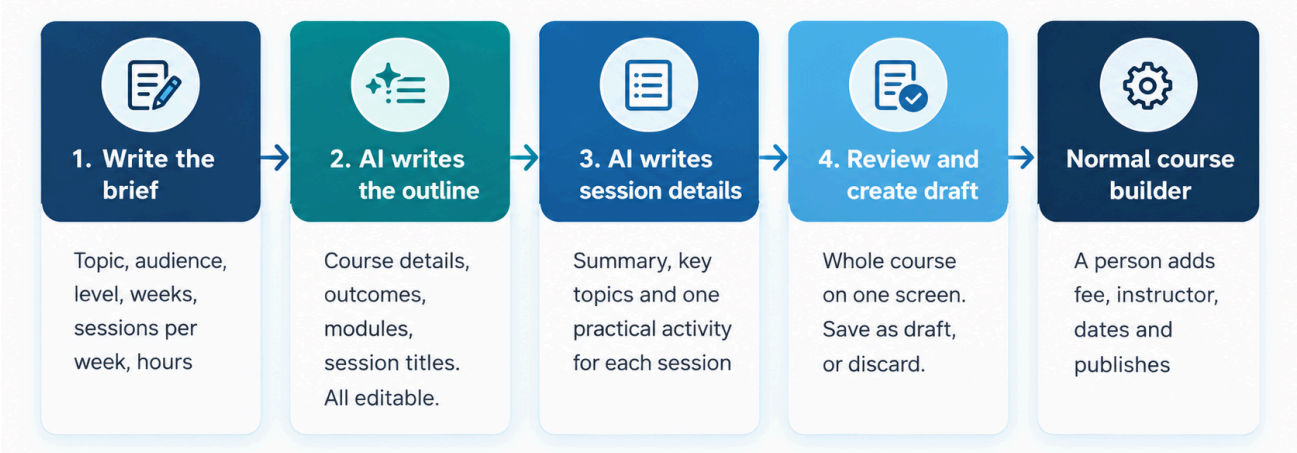
Three rules that never change

1. The AI writes the draft; a person decides. Nothing is saved until the coordinator accepts it. A generated course is always saved as a draft and is never published by itself.

2. The AI writes teaching content only. It never sets fees, instructors, venue, Zoom links, status or enrolments.

3. The feature reuses what the platform already has: the same AI connection, the same provider settings, and the same course and session records.

2. How it works for the user



The button "Generate with SDC Learn AI" sits on the courses page and opens a new page at **#/manage/courses/generate**.

Step 1: The brief

Field	Required	Notes
Course topic or working title	Yes	For example "Power BI for finance teams".
Target audience	Yes	Who the learners are and what they already know.
Level, program type, delivery	Yes	The lists come from Settings.
Number of weeks, sessions per week, hours per session	Yes	The form shows a live total of sessions and hours. It blocks a total above the limit (section 6.1).
Program and division	No	Same behaviour as the manual course form.
What learners should be able to do at the end	No	Free text.
Tools or software used	No	

Field	Required	Notes
Source material	No	Pasted syllabus or notes, up to 12,000 characters. Longer text is refused with a message, not cut. A notice says not to paste personal data or material SDC has no right to use.
Start date, class days, start time	No	Used by the platform to work out session dates (release 2). Never sent to the AI.

Step 2: The outline

One AI request returns: title, tagline, description, prerequisites, a suggested code and icon; 4 to 8 learning outcomes; the modules, each with a name, a one-line summary and its sessions (title and a one-line summary). The coordinator can edit every field, add, remove and reorder.

Step 3: The session details (release 2)

After the outline is accepted, the platform asks the AI for the details of each session: a summary, key topics, one practical activity and the outcomes it supports. Each request carries the full accepted outline, so the AI knows what comes before and after. Progress is shown. A failed part can be retried alone.

Step 4: Review and create

One screen shows the whole course as it will be saved, with totals and the notice "Drafted with SDC Learn AI. Review the content before publishing." **Create draft course** saves and opens the normal course builder. **Discard** throws the draft away.

3. Who can use it

No new permission is added. The button and the page appear only when all four are true:

- the user may create courses;
- the user may use AI;
- the "Course generator" switch in Settings → Features is on;
- AI is switched on for the platform.

With the roles the platform ships with, that means the **Coordinator**. Instructors can edit courses but cannot create them, so they do not see the generator. Learners and Accounts Officers do not see it.

4. What we build, and in what order

The feature is built in two releases, so that a useful part is working early and the larger part is built on real experience.

Stage	What it delivers	Why in this order
Release 1 Course outline	Brief → one AI request → editable outline (course details, outcomes, modules, session titles with one-line summaries) → review → draft course with all its sessions created.	One AI request, so it is fast and cannot be half finished. It already removes most of the form filling. Small enough to build and test quickly.
Release 2 Full session detail	Step 3: summary, topics and activity for every session; regenerate with an instruction; session dates worked out by the platform; outcome coverage check; "needs review" marks; the "Generate sessions with AI" button in the builder's Sessions tab.	Needs several AI requests per course and more rules (splitting, retry, marks). Built after release 1 has been tried on real courses.
Later phases	Detailed teaching content, educational videos, practical demonstrations (section 11).	Each needs new storage or new services. Each gets its own plan later.

5. Feature list

The IDs are the same as in the earlier documents. Must = required in that release. Should = expected. Could = nice to have.

ID	Requirement	Priority	Release
FR-CG-1	A "Generate with SDC Learn AI" button on the courses page, shown only to the people in section 3.	Must	1

ID	Requirement	Priority	Release
FR-CG-2	A brief form with the step 1 fields, input checks and a live total of sessions and hours.	Must	1
FR-CG-21	A brief over the size limit is refused on the form, before any AI request.	Must	1
FR-CG-3	The outline is generated in one AI request and shown in an editable form.	Must	1
FR-CG-6	Every AI answer is checked against a fixed format before it is shown. A bad answer is retried once, then the user sees an error.	Must	1
FR-CG-7	The platform enforces the number of sessions, modules and hours. It does not trust the AI's numbers.	Must	1
FR-CG-9	The AI never sets the fee, instructors, venue, Zoom details, status or enrolments.	Must	1
FR-CG-10	A generated course is always saved as a draft, even if the user may publish.	Must	1
FR-CG-11	A generated course is marked as AI-drafted. Provider, model, date and user are recorded on the course.	Must	1
FR-CG-12	The course builder shows an "AI draft" label until the course is published.	Must	1
FR-CG-14	Work in progress survives a page reload in the same browser tab, for the same user.	Should	1
FR-CG-16	Course generation has its own line in Settings → Integrations: provider, model and temperature.	Must	1

ID	Requirement	Priority	Release
FR-CG-17	An on/off switch "Course generator" in Settings → Features.	Must	1
FR-CG-18	Generation counts against the existing AI limits and monthly budget, and is listed under its own name in the AI usage report.	Must	1
FR-CG-23	Pressing save twice creates one course. A course name already in use is reported to the user. The user is told if the cloud save failed.	Must	1
FR-CG-15	Pasted source material, if given, is used as the main reference for the content.	Should	1
FR-CG-4	Session details are generated in separate requests. Each part shows progress and can be retried alone.	Must	2
FR-CG-5	The outline or a part can be regenerated with an extra instruction.	Must	2
FR-CG-8	The platform works out session dates and times from the start date, class days and start time.	Must	2
FR-CG-25	The chosen class days must match the number of sessions per week.	Must	2
FR-CG-13	The review screen checks that every outcome is taught in some session, and every session supports some outcome.	Should	2
FR-CG-22	If the outline is edited after details exist, the details that depend on it are marked "needs review". Regenerating never quietly overwrites a person's edits.	Should	2
FR-CG-19	A "Generate sessions with AI" button in the builder's Sessions tab, for a draft course that has outcomes and no sessions.	Could	2

ID	Requirement	Priority	Release
FR-CG-20	Draft assignments and quizzes from the course plan.	Later	Later phase

6. Rules the generator follows

6.1 Size limit

- At most **12 modules and 48 sessions** per course.
- A brief above 48 sessions is refused on the form. The course is never quietly shortened.

6.2 Size of each AI request

- Release 1 uses one request for the whole outline.
- Release 2 asks for details in groups of sessions, not per module. It starts at **6 sessions per request**; the final number comes from testing. A large module is split across requests.
- The platform numbers the session slots and the AI fills them, so a missing or extra session is easy to see.
- If an answer is cut off, that group is split in half and asked again. It is never repeated unchanged.
- Target time per request: under 30 seconds. The server limit is 60 seconds.

6.3 What the platform fixes, marks or refuses

Rule	Cases	What happens
Fix quietly (format only)	Code markers or extra text around the answer; spaces at the start or end; a line break or the character "|" inside a module name or outcome; unknown fields; module numbering; an icon outside the allowed list.	Fixed. The meaning does not change.
Keep and mark	Text longer than its limit; an outcome number that does not exist; a session with no outcome or an outcome with no session; an outcome that does not start with an	Shown whole with a mark on the review screen. Never cut, never rewritten.

Rule	Cases	What happens
	action word; a course name already in use.	
Refuse	A brief over the limit; an answer that is not valid, is empty or has a missing required field; a cut-off answer; more than 12 modules.	One automatic retry (a cut-off answer is split instead). Then an error for that part only. Never saved.

Never: invent a placeholder session, delete a session to fit the count, shorten text, or change something the person has already accepted.

Wrong number of sessions: after one retry, the outline is shown as returned, with missing slots marked "missing" and extra ones marked "extra". The person fixes it, and may save with a different count once they confirm it. Marks warn the user; they do not block saving.

6.4 Saving

- The course and all its sessions are written in one save.
- Just before saving, the generator loads the latest data, adds its new course and sessions, then saves.
- The save button is disabled after the first press.
- If the cloud save fails, the user is told the work is saved on this device only.
- If the write fails, the generator removes its own half-written records.
- Sessions are saved as visible. The course is a draft, so learners see nothing until a person publishes it.

6.5 Content rules written into the AI instructions

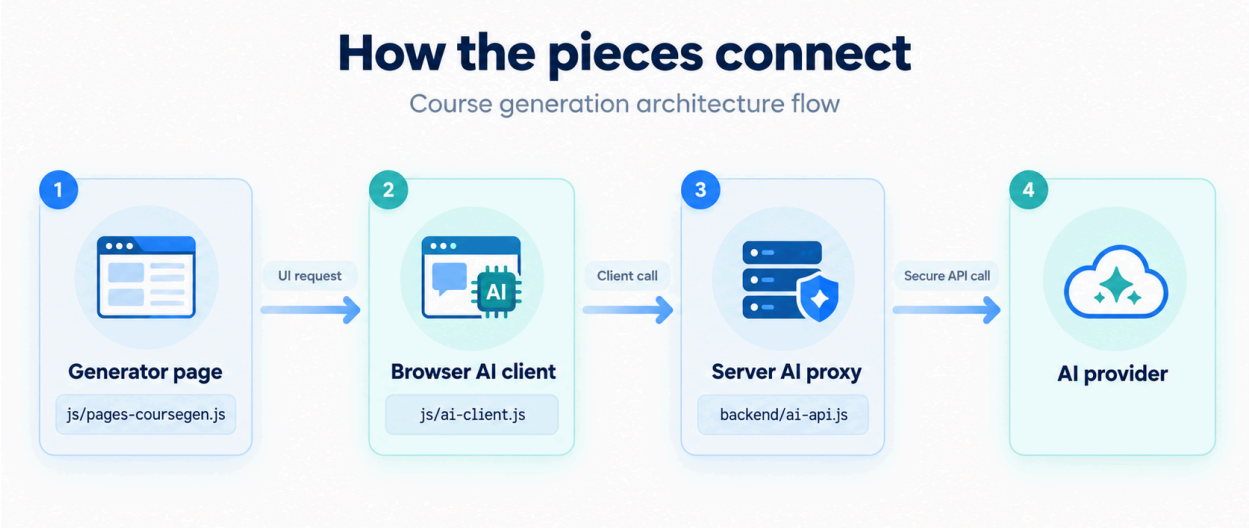
- Outcomes can be measured and start with an action word (build, analyse, configure). No "understand" or "learn about".
- The course gets harder as it goes, and each module builds on the one before.
- Every session has one practical activity that fits the tools named in the brief (release 2).
- The last module includes a project or assessment that brings the outcomes together.
- Each session lists which outcomes it supports.
- Every request carries a short description of SDC Karachi and who it teaches (working professionals and job seekers; practical, job-focused training), plus the chosen program and division.
- The outline request also carries the titles of existing courses and one published course as an example of the expected style.
- The AI must answer in a fixed data format (JSON). Temperature 0.4, changeable in Settings.

6.6 When something goes wrong

Situation	What the user sees
No AI provider is set up	The existing message asking for a provider in Settings → Integrations.
AI is off, or the user has no permission	The button is hidden. A direct link shows the normal "no access" page.
The monthly budget is used up	The existing budget message. The draft so far is kept.
Too many requests in an hour	The existing limit message, with a retry button.
The AI answer is broken or cut off	Section 6.3, then "Could not generate this part. Try again." for that part only.
The provider fails	The platform tries the next configured provider. If that also fails, an error with a retry button.
The page is reloaded part-way	The draft is restored.

7. Technical design

7.1



1. The page calls SDCAI.call('courseGenerate', …).
2. That sends **POST /api/ai/complete**. The server applies the existing limits and budget and calls the provider. Provider keys stay on the server.
3. The answer comes back to the browser, is checked against the fixed format (section 6.3) and shown for review.
4. On "Create draft course" the records go through the normal store functions in js/core.js and the normal cloud sync.

7.2 Changes needed in the AI connection

The existing AI features send short questions. Course generation sends long, structured ones, so the connection needs four small additions. They also help every other AI feature.

Addition	Today	Needed
Fixed answer format (JSON mode)	Not passed to any provider.	Passed from the page through the client to the provider.
Reason the answer stopped	Thrown away, so a cut-off answer looks like a broken one.	Returned to the page, so a cut-off part can be split.

Addition	Today	Needed
Answer size	2,048 tokens unless the page asks for more.	The generator asks for a larger size, with a fixed upper limit.
Timeout	None. The server stops everything at 60 seconds.	A timeout below 60 seconds, so there is time for the fallback provider and a clear error.
Error codes	The page cannot tell the errors apart.	Separate codes for limit reached, budget reached and provider failed.

7.3 Data that is stored

No new type of record. Two new optional fields on the course: **aiGenerated** (true or false) and **aiMeta** (provider, model, date, user). New default settings appear without a data migration.

Course field	Where the value comes from
Title, tagline, description, prerequisites, outcomes, modules	The AI, then edited by the reviewer.
Code, icon	Suggested by the AI. The icon must be one from the existing list.
Web address name (slug)	The platform, from the title. A clash is reported to the user.
Level, program type, delivery, program, division	The brief.
Duration, start date, end date, schedule	The platform.
Status	Always draft.
Fee, venue, Zoom, help link	Left empty for a person.

Sessions use the existing fields: title, moduleName, summary, duration, date, time, order,

published, resources, videoUrl.

- Sessions belong to a module by its exact name, so the generator writes the same module name on the course and on each session.
- Release 1 writes a one-line summary. Release 2 writes summary, key topics and activity into the summary field, with line breaks kept when it is shown.
- If a user whose course access is not "all" creates a course, the platform also creates a first batch for them, exactly as the manual form does.

7.4 Files that change

File	Change
js/pages-coursegen.js (new)	The generator page and its route.
js/coursegen-logic.js (new)	Pure logic that can be tested without a page: format check, the fix/mark/refuse rule, size limit, slot numbering, date calculation, coverage check.
app.html, package.json	Load the two new scripts and add them to the check script.
js/pages-manage.js	The button, the "AI draft" label, and one shared "create course" function used by both the form and the generator. In release 2, the Sessions tab button.
js/core.js, js/pages-ops.js	Default settings, the feature switch and labels in Settings.
js/ai-client.js	Pass the fixed-format setting; return error codes the page can tell apart.
backend/ai-api.js	The capability name and the additions in section 7.2.
css/app.css	Styles for the new screens.
tests, FEATURES.md, docs/IMPLEMENTATION.md	Tests and documentation.

Not changed: js/cloud-config.js, js/seed.js, the roles, the learner pages.

7.5 Expected AI use

Item	Estimate (to be measured in the trial)
Requests per course, release 1	1, plus 1 for each retry.
Requests per course, release 2	1 for the outline plus 1 per group of 6 sessions. About 5 for 24 sessions, about 9 for 48. Each retry or regenerate adds 1.
Tokens per course, release 1	About 3,600 for a 6-module course; about 7,300 at the maximum size with pasted material.
Tokens per course, release 2	About 25,000 to 33,000 for 6 modules and 24 sessions; up to about 110,000 at the maximum size with pasted material.
Existing limits	60 requests per user per hour and 5,000 requests per month, shared with the other AI features.

8. Build steps

Release 1: course outline

Step	Work	Files
1.1	AI connection additions: fixed format, stop reason, answer size limit, timeout, error codes. Check that the existing AI features still work.	backend/ai-api.js, js/ai-client.js
1.2	Settings: the "courseGenerate" capability, its line in Integrations, the "Course generator" switch, labels.	js/core.js, js/pages-ops.js
1.3	Pure logic with tests: format check, fix/mark/refuse rule, size limit, slot numbering.	js/coursegen-logic.js, tests
1.4	Generator page: route, brief form with live totals and the size	js/pages-coursegen.js,

Step	Work	Files
	block.	app.html, css/app.css
1.5	Outline request, the AI instructions (section 6.5), and the editable outline with marks.	js/pages-coursegen.js
1.6	Review screen and "Create draft course": shared create function, one save, duplicate and cloud-save feedback, AI mark on the course.	js/pages-manage.js, js/pages-coursegen.js
1.7	Button on the courses page, "AI draft" label in the builder, draft kept across a page reload.	js/pages-manage.js
1.8	Run npm run check and npm test. Update FEATURES.md and docs/IMPLEMENTATION.md. Push to the Noman branch.	

Release 2: full session detail

Step	Work
2.1	Detail requests in groups of sessions, with progress, retry per part and splitting of cut-off parts.
2.2	Regenerate the outline or a part with an instruction; confirmation that names the edited fields that would be replaced.
2.3	Session dates from start date, class days and start time; class days must match sessions per week.
2.4	Outcome coverage check and "needs review" marks on the review screen.
2.5	Session summary shown with line breaks kept.
2.6	"Generate sessions with AI" button in the builder's Sessions tab for a draft course with outcomes and no sessions.

Step	Work
2.7	Tests, documentation, push to the Noman branch.

9. How we know it is finished

No.	Test	Release
1	A user who may create courses and use AI, with both switches on, sees the button and can open the page. Any other user does not see the button and gets the normal "no access" page from a direct link.	1
2	A brief over the limit cannot be submitted, and the AI usage count does not change.	1
3	One AI request returns the outline; the usage count rises by exactly one; every outline field can be edited, added, removed and reordered before saving.	1
4	Every case in section 6.3 behaves as written. No test ends with a session added, deleted or shortened by the platform.	1
5	The saved session count and hours equal the brief, or the count the person confirmed.	1
6	After generating: the course is a draft; fee, venue, Zoom and help link are empty; no enrolment was created; no instructor was chosen by the AI.	1
7	The course carries the AI mark with provider, model, date and user. The "AI draft" label shows until the course is published.	1
8	After a page reload the draft is restored for the same user.	1
9	"Course generation" appears in Settings → Integrations, and the model set there is the model named on the saved course.	1

No.	Test	Release
10	Usage appears under the feature's own name in the AI usage report and counts against the monthly budget.	1
11	Pressing the save button twice creates one course. A name already in use is reported, not silently changed.	1
12	With the cloud unreachable, the user is told the work is saved on this device only.	1
13	The generated course and its sessions are not shown on any learner screen before publishing.	1
14	The existing AI features (tutor, summarizer, quizzes, grading) still work after the AI connection changes.	1
15	npm run check and npm test pass, and FEATURES.md describes the feature.	1 and 2
16	A failed detail part can be retried alone; the other parts and all hand edits are still there afterwards.	2
17	Regenerating over edited fields asks for confirmation and names the fields.	2
18	With a start date and class days, every session has the right date in order. With none, dates are empty. Class days that do not match sessions per week are refused.	2
19	An outcome with no session and a session with no outcome are both marked on the review screen.	2
20	A coordinator goes from an empty brief to a saved draft course with all session details without leaving the generator.	2

10. Trying it on real courses

After release 1 is working, it is tried on **three to five real courses** SDC is about to create,

before release 2 is built.

- One course is also built by hand and timed, as the comparison.
- The instructor who will teach each course scores the draft: how many sessions had to be rewritten, any factual errors, minutes spent reviewing, and whether they would use it again.
- Time per request and tokens per course are recorded, to replace the estimates in section 7.5.
- The result decides the details of release 2: the number of sessions per request and how much detail instructors actually want.

11. Later phases

These are the direction after course generation. They are not designed yet. Each needs its own plan before any build.

Phase	What it would deliver	What it needs first
Teaching content	Detailed lesson notes, worked examples, practical exercises, an instructor guide, a narration script and slide outline, and assessment plans. All editable, with a history of versions.	Results from course generation. A place to store long content outside the single platform data file.
Educational videos	Start simple: the AI drafts a script and slide outline, the instructor records and pastes the link (the session page already plays links). Later: automatic narrated slides with an AI voice and captions.	For the simple path: nothing new. For automatic videos: a speech service, a slide renderer, a video encoder, background jobs and media storage. None exists today.
Practical demonstrations	Screen recordings made and checked by an instructor first. Automatic recording only after separate research.	Nothing for recordings added as links. Media storage if files are uploaded.

Language stays English only. Assignments and quizzes keep belonging to batches: an AI assessment plan becomes a live assignment or quiz only when an instructor creates it for a batch.

12. Not included

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  Publishing, enrolling learners, or issuing anything automatically.
- Fees, instructors, venues or Zoom details.
- Videos, slides, recordings or file uploads (later phases).
- Languages other than English.
- Rewriting a course that already has learners, submissions or attendance.
- Checking that content is technically correct. The reviewer stays responsible.

13. Choices made in this plan

The earlier documents left these open. This plan picks one answer for each so that work can start. Change any of them before building begins if the team disagrees.

Question	Choice in this plan
Who can generate courses?	Users who may create courses and use AI (the Coordinator role as shipped).
What comes first?	Release 1 is the outline with session titles. Full session detail is release 2.
What can the AI work from?	The brief and pasted text. File upload later.
Which provider and model?	The one set for "Course generation" in Settings → Integrations. The model used is recorded on the course.
How is a course published?	Saved as a draft only. A person publishes.
How big can a course be?	12 modules and 48 sessions. To confirm: is this enough for SDC's longest course?
Are generated sessions visible or hidden?	Visible. The draft course hides them from learners until it is published.
When does the "AI draft" label	When the course is published.

Question	Choice in this plan
go away?	
May the saved session count differ from the brief?	Yes, after the person confirms it. Marks warn; they do not block saving.
Where do topics and activity go (release 2)?	In the session summary, with line breaks kept on display. No new session fields.
Is the switch on or off at first?	Off. It is turned on when the trial in section 10 starts.

14. Project rules for this work

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  Push only to the Noman branch. Never push to main and never open a pull request into main.
- Do not change js/cloud-config.js.
- AI provider keys stay on the server.
- Never run tests/e2e.browser.js against live data.
- npm run check and npm test must pass before pushing.
- Update FEATURES.md in the same change as any feature, route, permission or setting.
- Plain HTML, CSS and JavaScript only. No new framework or build tool.